

SHETH L.U.J. AND SIR M.V. COLLEGE

Generating correlation matrices using cor() (R).

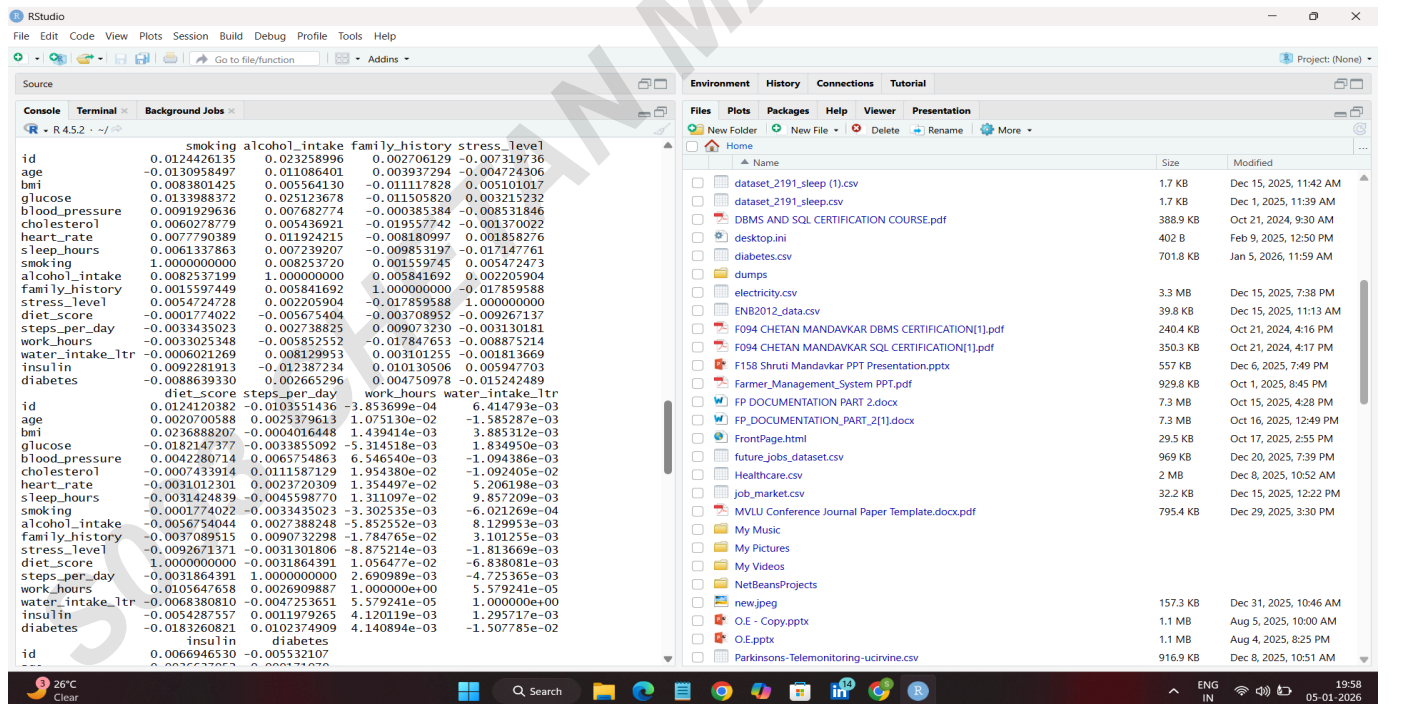
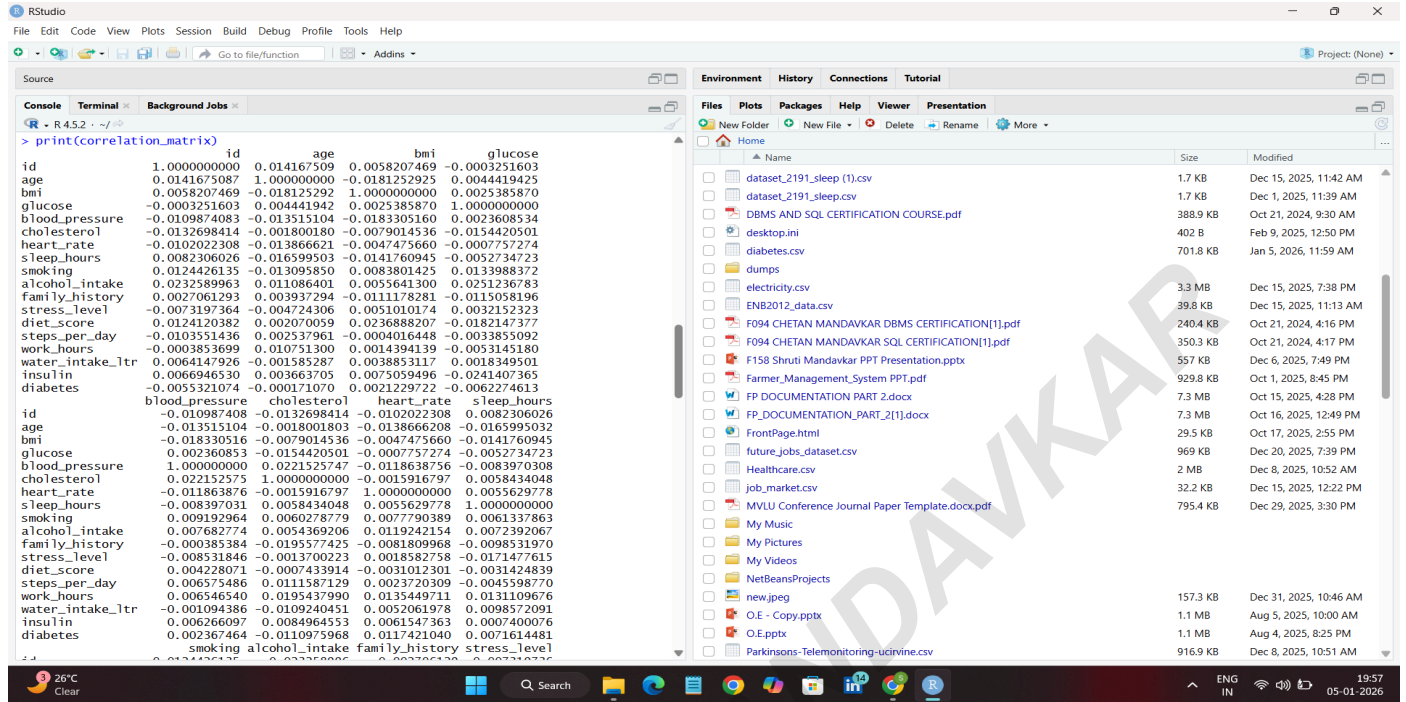
```
RStudio
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Source
Console Terminal Background Jobs
> diabetes_data <- read.csv("diabetes.csv")
> head(diabetes_data)
  id age gender  bmi glucose blood_pressure cholesterol heart_rate
1 1 69 Male 23.30 170 137 139 66
2 2 32 Male 25.00 184 177 250 75
3 3 89 Female 28.57 87 164 225 102
4 4 78 Male 15.85 96 113 158 112
5 5 38 Female 35.74 171 122 193 109
6 6 41 Male 26.24 98 107 209 60
  sleep_hours physical_activity smoking alcohol_intake family_history
1 6.4 High 1 1 1
2 6.4 Medium 0 1 1
3 7.5 Medium 1 0 0
4 6.9 Low 0 1 1
5 7.4 Medium 0 1 0
6 9.9 Medium 1 0 1
  stress_level diet_score steps_per_day work_hours water_intake_ltr insulin
1 8 3 8760 6 3.6 33
2 4 7 7682 11 4.1 36
3 6 5 15025 8 1.4 46
4 2 7 4645 4 4.4 171
5 6 9 7862 13 3.9 235
6 2 10 4849 12 3.1 30
diabetes
1 0
2 0
3 1
4 1
5 0
6 0
> str(diabetes_data)
'data.frame': 10000 obs. of 20 variables:
 $ id : int 1 2 3 4 5 6 7 8 9 10 ...
 $ age : int 69 32 89 78 38 41 20 39 70 19 ...
 $ gender : chr "Male" "Male" "Female" "Male" ...
 $ bmi : num 23.3 25.0 28.6 15.8 35.7 ...
 $ glucose : int 170 184 87 96 171 98 98 171 160 74 ...
 $ blood_pressure : int 137 177 164 113 122 107 135 126 114 157 ...
 $ cholesterol : int 139 250 225 158 193 209 257 287 260 163 ...

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> head(diabetes_data)
  id age gender  bmi glucose blood_pressure cholesterol heart_rate
1 1 69 Male 23.30 170 137 139 66
2 2 32 Male 25.00 184 177 250 75
3 3 89 Female 28.57 87 164 225 102
4 4 78 Male 15.85 96 113 158 112
5 5 38 Female 35.74 171 122 193 109
6 6 41 Male 26.24 98 107 209 60
  sleep_hours physical_activity smoking alcohol_intake family_history
1 6.4 High 1 1 1
2 6.4 Medium 0 1 1
3 7.5 Medium 1 0 0
4 6.9 Low 0 1 1
5 7.4 Medium 0 1 0
6 9.9 Medium 1 0 1
  stress_level diet_score steps_per_day work_hours water_intake_ltr insulin
1 8 3 8760 6 3.6 33
2 4 7 7682 11 4.1 36
3 6 5 15025 8 1.4 46
4 2 7 4645 4 4.4 171
5 6 9 7862 13 3.9 235
6 2 10 4849 12 3.1 30
diabetes
1 0
2 0
3 1
4 1
5 0
6 0
> str(diabetes_data)
'data.frame': 10000 obs. of 20 variables:
 $ id : int 1 2 3 4 5 6 7 8 9 10 ...
 $ age : int 69 32 89 78 38 41 20 39 70 19 ...
 $ gender : chr "Male" "Male" "Female" "Male" ...
 $ bmi : num 23.3 25.0 28.6 15.8 35.7 ...
 $ glucose : int 170 184 87 96 171 98 98 171 160 74 ...
 $ blood_pressure : int 137 177 164 113 122 107 135 126 114 157 ...
 $ cholesterol : int 139 250 225 158 193 209 257 287 260 163 ...
 $ sleep_hours : num 6.4 6.4 7.5 6.9 7.4 9.9 8.7 5 6.7 4.6 ...
 $ physical_activity : chr "High" "Medium" "Medium" "Low" ...
 $ smoking : int 1 0 1 0 1 0 0 1 0 ...
 $ alcohol_intake : int 1 1 0 1 0 0 0 1 1 ...
 $ family_history : int 1 1 0 1 0 1 0 1 0 ...
 $ stress_level : int 8 4 6 2 6 2 1 5 4 4 ...
 $ diet_score : int 3 7 5 7 9 10 1 6 9 1 ...
 $ steps_per_day : int 8760 7682 15025 4645 7862 4849 11604 7398 3735 16432 ...
 $ work_hours : int 6 11 8 4 13 12 8 9 5 13 ...
 $ water_intake_ltr : num 3.6 4.1 1.4 4.4 3.9 3.1 4.7 4.1 3 4.5 ...
 $ insulin : int 33 36 46 171 235 30 26 96 21 231 ...
 $ diabetes : int 0 0 1 1 0 0 0 0 0 ...
> numeric_data <- diabetes_data[sapply(diabetes_data, is.numeric)]
> head(numeric_data)
  id age  bmi glucose blood_pressure cholesterol heart_rate sleep_hours
1 1 69 23.30 170 137 139 66 6.4
2 2 32 25.00 184 177 250 75 6.4
3 3 89 28.57 87 164 225 102 7.5
4 4 78 15.85 96 113 158 112 6.9
5 5 38 35.74 171 122 193 109 7.4
6 6 41 26.24 98 107 209 60 9.9
  smoking alcohol_intake family_history stress_level diet_score
1 1 1 1 1 1 1 1 1 1
2 0 1 1 1 1 1 1 1 1
3 1 1 1 1 1 1 1 1 1
4 0 1 1 1 1 1 1 1 1
5 0 1 1 1 1 1 1 1 1
6 1 1 1 1 1 1 1 1 1
  steps_per_day work_hours water_intake_ltr insulin diabetes
1 8760 6 3.6 33 0
2 7682 11 4.1 36 0
3 15025 8 1.4 46 1
4 4645 4 4.4 171 1
5 7862 13 3.9 235 0
6 4849 12 3.1 30 0
> correlation_matrix <- cor(numeric_data)
> print(correlation_matrix)
      id age  bmi glucose
id      1 0.0000000 0.0000000 0.0000000 0.0000000
age     0 1.0000000 0.0000000 0.0000000 0.0000000
bmi     0 0.0000000 1.0000000 0.0000000 0.0000000
glucose 0 0.0000000 0.0000000 1.0000000 0.0000000
blood_pressure 0 0.0000000 0.0000000 0.0000000 1.0000000
cholesterol 0 0.0000000 0.0000000 0.0000000 0.0000000 1.0000000
heart_rate 0 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 1.0000000
sleep_hours 0 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 1.0000000
physical_activity 0 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 1.0000000
smoking 0 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 1.0000000
alcohol_intake 0 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 1.0000000
family_history 0 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 1.0000000
stress_level 0 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 1.0000000
diet_score 0 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 1.0000000
steps_per_day 0 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 1.0000000
work_hours 0 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 1.0000000
water_intake_ltr 0 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 1.0000000
insulin 0 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 1.0000000
diabetes 0 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 1.0000000
```

NAME:- CHETAN MANDAVKAR
ROLL NO. S093
SUBJECT:- Data Analysis With SAS / SPSS / R

SHETH L.U.J. AND SIR M.V. COLLEGE



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```
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Go to file/function Addins
Source
Console Terminal Background Jobs
R - R4.5.2 - ~/
diabetes
id 0.0066946530 -0.00532107
age 0.0036637052 -0.000171070
bmi 0.0075059496 0.002122972
glucose -0.0241407365 -0.006227461
blood_pressure 0.0062660966 0.002367464
cholesterol 0.0084964553 -0.011097597
heart_rate 0.0061547363 0.011742104
sleep_hours 0.0007400076 0.007161448
smoking 0.0092281913 -0.008863933
alcohol_intake -0.0123872340 -0.002665296
family_history 0.0101305057 0.004750978
stress_level 0.0059477029 -0.015242489
diet_score -0.0054287557 -0.018326082
steps_per_day 0.0011979265 0.010237491
work_hours 0.0041201189 0.004140894
water_intake_ltr 0.0012957173 -0.015077853
insulin 1.0000000000 -0.011514265
diabetes -0.0115142646 1.0000000000
> rounded_correlation <- round(correlation_matrix, 2)
> print(rounded_correlation)
id age bmi glucose blood_pressure cholesterol
id 1.00 0.01 0.01 0.00 -0.01 -0.01
age 0.01 1.00 -0.02 0.00 -0.01 0.00
bmi 0.01 -0.02 1.00 0.00 -0.02 -0.01
glucose 0.00 0.00 0.00 1.00 0.00 -0.02
blood_pressure -0.01 -0.01 -0.02 0.00 1.00 0.02
cholesterol -0.01 0.00 -0.01 -0.02 0.02 1.00
heart_rate -0.01 -0.01 0.00 0.00 -0.01 0.00
sleep_hours 0.01 -0.02 -0.01 -0.01 -0.01 0.01
smoking 0.01 -0.01 0.01 0.01 0.01 0.01
alcohol_intake 0.02 0.01 0.01 0.03 0.01 0.01
family_history 0.00 0.00 -0.01 -0.01 0.00 -0.02
stress_level -0.01 0.00 0.01 0.00 0.00 0.00
diet_score 0.01 0.00 0.02 -0.02 0.00 0.00
steps_per_day -0.01 0.00 0.00 0.00 0.01 0.01
work_hours 0.00 0.01 0.00 -0.01 0.01 0.02
water_intake_ltr 0.01 0.00 0.00 0.00 0.00 -0.01
insulin 0.01 0.00 0.01 -0.02 0.01 0.01
diabetes -0.01 0.00 0.00 -0.01 0.00 -0.01
```

```
RStudio
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Go to file/function Addins
Source
Console Terminal Background Jobs
R - R4.5.2 - ~/
family_history stress_level diet_score steps_per_day
id 0.00 -0.01 0.01 -0.01
age 0.00 0.00 0.00 0.00
bmi -0.01 0.01 0.02 0.00
glucose -0.01 0.00 -0.02 0.00
blood_pressure 0.00 -0.01 0.00 0.01
cholesterol -0.02 0.00 0.00 0.01
heart_rate -0.01 0.00 0.00 0.00
sleep_hours -0.01 -0.02 0.00 0.00
smoking 0.00 0.01 0.00 0.00
alcohol_intake 0.01 0.00 -0.01 0.00
family_history 1.00 -0.02 0.00 0.01
stress_level -0.02 1.00 -0.01 0.00
diet_score 0.00 -0.01 -0.01 0.00
steps_per_day 0.01 0.00 0.00 1.00
work_hours -0.02 -0.01 0.01 0.00
water_intake_ltr 0.00 0.00 -0.01 0.00
insulin 0.01 0.01 -0.01 0.00
diabetes 0.00 -0.02 -0.01 0.01
work_hours water_intake_ltr insulin diabetes
id 0.00 0.01 0.01 -0.01
age 0.01 0.00 0.00 0.00
bmi 0.00 0.01 0.01 0.00
glucose -0.01 0.00 -0.02 -0.01
blood_pressure 0.00 0.01 0.00 0.00
cholesterol 0.02 -0.01 0.01 -0.01
heart_rate 0.01 0.01 0.01 0.01
sleep_hours 0.01 0.01 0.00 0.01
smoking 0.00 0.00 0.01 -0.01
alcohol_intake -0.01 -0.01 -0.01 0.00
family_history 0.00 0.00 0.01 0.00
stress_level -0.01 0.00 0.01 -0.02
diet_score 0.01 -0.01 -0.01 -0.02
steps_per_day 0.00 0.00 0.00 0.01
work_hours 0.00 0.00 0.00 0.00
water_intake_ltr 0.00 1.00 0.00 -0.02
insulin 0.00 0.00 1.00 -0.01
diabetes 0.00 -0.02 -0.01 1.00
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